

Physical Activity Modeling Summary

1. Data Preparation

The ESS6 dataset was cleaned to remove invalid responses for variables including happiness, income, age, and gender. The outcome variable (physical activity) was transformed into both an ordinal factor (**Inactive** < **Active**) and a binary factor. Gender was labeled as "Male" or "Female", and invalid codes were excluded. The final dataset was ready for regression modeling.

2. Ordinal Logistic Regression

The ordinal logistic regression model examined the ordered levels of physical activity across predictors:

- **Age** had a small, statistically insignificant positive effect.
- **Gender** showed that females are more likely to be active than males ($p < 0.001$).
- **Happiness** was the strongest predictor with a large positive effect ($p < 2e-88$).
- **Income** had a very small and statistically insignificant effect.

Country Effects (Reference = Ireland)

- Germany had the highest positive coefficient, suggesting Germans are significantly more likely to be active than Irish respondents.
- Israel had the most negative coefficient, indicating much lower odds of physical activity.
- Countries such as Spain, Portugal, and Switzerland also showed significantly lower activity.

3. Odds Ratios from Ordinal Model

- **Happiness** increased the odds of being more active by approximately 10% per unit increase.

- **Germany** had an odds ratio of ~2.85, indicating Germans were almost 3 times more likely to be active than Irish.
- **Negative ORs** (e.g., Israel ~0.117) suggested lower likelihood compared to Ireland.

4. Binary Logistic Mixed Model (GLMM)

The binary GLMM included a random intercept for country:

- **Age:** Slightly decreases the odds of activity.
- **Gender:** Not statistically significant.
- **Happiness:** Strong positive predictor.
- **Income:** Small but significant positive effect.

The intercept reflects the baseline odds of activity in Ireland.

5. Model Comparison

- Ordinal model AIC: 155583.8
- Binary GLMM AIC: 42239.3

The GLMM is a better fit due to lower AIC.

6. Quadratic Age Term

A quadratic term for age significantly improved the model (Chi-squared = 197.36, $p < 2.2e-16$).

- Probability of being active increases until middle age (~45-50), then declines.

7. Effect Visualizations

- **Age:** Arch-shaped curve—physical activity peaks in midlife.

- **Gender:** Similar probability (~84%) for both genders.
- **Happiness:** Strong linear increase in activity likelihood.
- **Income:** Nearly flat, confirming small effect.

8. Gender-Income Interaction

Adding an interaction between gender and income did not improve model fit ($p = 0.5716$). There is no evidence that gender modifies the effect of income on physical activity.

9. Country-Specific Random Effects

- Visualized country-level deviations.
- Countries like Slovakia and Bulgaria showed higher-than-average activity probabilities.
- Countries like Israel and Spain showed much lower probabilities.

Intraclass Correlation Coefficient (ICC) = 0.093 → About 9.3% of variance in activity is due to between-country differences.

10. Sensitivity Analysis

Excluding countries with fewer than 100 observations did not change the fixed effect estimates. The model is robust and results are consistent across sample sizes.

Final Summary

- **Happiness** is the most influential predictor.
- **Age** follows a quadratic relationship.
- **Gender** is not significant when adjusting for other factors.
- **Income** has a small but reliable effect.
- **Country-level variation** is substantial (ICC = 9.3%).

- **Results are stable** under different model assumptions and data subsets.

This modeling framework provides strong evidence of psychological and demographic influences on physical activity across Europe.